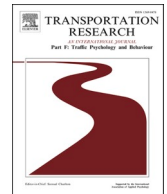




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Effect of multimodal takeover request issued through A-pillar LED light, earcon, speech message, and haptic seat in conditionally automated driving

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ABSTRACT

The driver of a conditionally automated vehicle equivalent to level 3 of the SAE is obligated to accept a takeover request (TOR) issued by the vehicle. Considerable research has been conducted on the TOR, especially in terms of the effectiveness of multimodal methods. Therefore, in this study, the effectiveness of various multimodalities was compared and analyzed. Thirty-six volunteers were recruited to compare the effects of the multimodalities, and vehicle and physiological data were obtained using a driving simulator. Eight combinations of TOR warnings, including those implemented through LED lights on the A-pillar, earcon, speech message, or vibrations in the back support and seat pan, were analyzed to clarify the corresponding effects. When the LED lights were implemented on the A-pillar, the driver reaction was faster ($p = 0.022$) and steering deviation was larger ($p = 0.024$) than those in the case in which no LED lights were implemented. The speech message resulted in a larger steering deviation than that in the case of the earcon ($p = 0.044$). When vibrations were provided through the haptic seat, the reaction time ($p < 0.001$) was faster, and the steering deviation ($p = 0.001$) was larger in the presence of vibrations in the haptic seat than no vibration. An interaction effect was noted between the visual and auditory modalities; notably, the earcon resulted in a small steering deviation and skin conductance response amplitude (SCR amplitude) when implemented with LED lights on the A-pillar, whereas the speech message led to a small steering deviation and SCR amplitude without the LED lights. In the design of a multimodal warning to be used to issue a TOR, the effects of each individual modality and corresponding interaction effects must be considered. These effects must be evaluated through application to various takeover situations.

1. Introduction

In conditionally automated vehicles classified as Level 3 autonomous cars according to the SAE J3016 standard, the automated driving mode may be deactivated by the driver or vehicle (SAE J3016, 2018). In vehicles with this level of automation, drivers are not obligated to keep their eyes on the road. However, if the vehicle requests the driver to take over control, through a takeover request (TOR), the driver should implement the dynamic driving task fallback. Therefore, in Level 3 vehicles, interactions between the driver and vehicle are crucial to ensure road safety. Consequently, original equipment manufacturers and academic societies are conducting

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Nomenclature

TOR	takeover requests
V	visual-modality warning
A	auditory-modality warning
H	haptic-modality warning
RT	reaction time
PT	perception time
MA	maximum acceleration
SWD	steering wheel angle deviation
A	A-pillar LED light & cluster
C	cluster only
E	earcon
S	speech message
H	haptic seat
N	none

considerable research on various types of warnings to ensure an effective takeover from the perspective of human–machine interface (HMI). In addition, international organizations, such as the UNECE World Forum for Harmonization of Vehicle Regulations (UNECE WP29), are actively hosting gatherings of representatives from multiple countries to enact relevant regulations for takeover scenarios.

[Brandenburg and Roche \(2020\)](#) evaluated the takeover performance depending on the repetition of the takeover process, behavioral change, and takeover step(s). When drivers were subjected to repeated takeovers, a stronger deceleration was observed. An unfamiliar situation requiring control takeover caused the driver to put in additional effort. A two-step takeover led to higher takeover times, higher time-to-collision, less steering, and lower perceived criticality ratios. [Naujoks et al. \(2019\)](#) summarized the recommendations for the design of a visual-auditory and visual-vibrotactile HMI of automated vehicles based on the applicable standards, in-vehicle interface design guidelines, and empirical research. [Morales-Alvarez et al. \(2020\)](#) described various aspects related to TOR, such as takeover situations, TOR framework, warnings, and regulations, and discussed future research directions.

Many researchers have reported that compared to single modalities, multimodal combinations are more effective in a takeover situation. In particular, [Bazilinskyy et al. \(2015\)](#) compared the success rates of a takeover when visual and haptic warnings were provided and reported that multimodalities were more successful in ensuring a takeover than single modalities. [Petermeijer et al. \(2017\)](#) compared the driver response time to a TOR in cases involving combinations of auditory and haptic signals and noted that the bimodality warning (auditory + haptic) initiated a faster response than that achieved through single-modality warnings. [Salminen et al. \(2019\)](#) analyzed the effects of unimodal (visual, auditory, or haptic) and multimodal warnings and highlighted that multimodal warnings led to a faster response compared to a unimodal signal. [Yun and Yang \(2020\)](#) used a combination of visual, auditory, and haptic modalities for a TOR and observed that the visual-auditory-haptic combination led to a faster reaction compared to visual, auditory, visual-auditory, and auditory-haptic combinations; in addition, the time required to change lanes was smaller than that in the case of visual and visual-auditory signals. Moreover, this combination was associated with a skin conductance response onset time faster than that for visual and visual-auditory feedbacks. The visual-only warning was inferior to other warning combinations in terms of the takeover performance. Overall, the visual-auditory-haptic combination corresponded to the highest takeover performance, whereas the visual warning was required to be combined either with an auditory signal or a haptic warning to be effective.

In this study, we extend the previous study conducted by [Yun and Yang \(2020\)](#), in which it was concluded that the visual-auditory-haptic multimodal warning was the most effective for implementing a takeover process. In this study, LED lights on the A-pillar were introduced as the visual warning modality, earcon and speech messages as the auditory warning modality, and vibration through the seat as the haptic warning modality. Subsequently, the warnings were combined in multimodal scenarios, and the corresponding effects were compared by conducting a driving simulator experiment.

The findings are expected to facilitate take over research as well, owing to the following aspects. First, we analyzed the performance of TOR warnings in a driving simulator by using the human-in-the-loop approach from the perspective of the driver, owing to which, the results can be compared with those of TOR studies (the implications are specified in [Section 2](#)). In addition, the effectiveness of multimodal warnings has been demonstrated internationally, given that the recommendation for the use of multimodal combinations for TOR warnings is consistent with the results of takeover studies, such as those of [Naujoks et al. \(2019\)](#) and [Morales-Alvarez et al. \(2020\)](#). However, this study is not a replication of TOR research and its novelty lies in the consideration of more diverse and detailed multimodal warning interfaces (e.g., A-pillar LED light, earcon, and seat vibration) and comparison of the design of each warning system. Therefore, the present findings are expected to be of significance in TOR studies.

2. Literature review

The takeover performance varies with the design of the modality. The design of a visual modality warning is based on an image (icon, picture, text, color, size, etc.) and its location. In the previous study, an icon was placed in the same location as a cluster and head-up display (HUD). [Melcher et al. \(2015\)](#), [Kim and Yang \(2020\)](#), and [Kraft et al. \(2020\)](#) suggested that as a medium to initiate

takeover in a system failure situation, the HUD was preferable over the cluster, and a dynamic warning was more appropriate than a static one. Borojeni et al. (2016) implemented an ambient light on the steering wheel through an LED strip. Çapalar and Olaverri-Monreal (2017) demonstrated the potential of visual stimulation with an ambient light system to draw the attention of the driver to promptly respond to a TOR.

Using only a visual warning as the takeover warning was not recommended because a considerable time was required for the driver to become aware of the takeover situation and operate the vehicle. However, because visual warnings include detailed information such as the road situations, its use in combination with an auditory or haptic warning is of significance (Yun and Yang, 2020).

An auditory modality warning results in a faster driver response than a visual modality (Woodworth & Schlosberg, 1954). In addition, an auditory modality reduces the burden of visual information processing (Wickens et al., 2002). Richie et al. (2018) analyzed the existing guidelines and the literature to provide guidelines for auditory modality warnings suitable to initiate takeover. The types of auditory warnings were experimentally compared, and it was noted that the use of earcon and speech messages led to a faster takeover and higher satisfaction than those when spearcon was used (Jeon, 2019). Forster et al. (2017) compared the effects of a generic condition (consistent repetition of a certain tone) and the combination of generic and speech messages in a simulator experiment and indicated that the volunteers preferred the warning with the speech message.

In this regard, a haptic modality may lead to a higher cognition of the situations. When establishing a haptic warning, two factors must be considered, specifically, the location (steering wheel, pedal, seat belt, driver seat, etc.) and delivery mode (vibration, jerk, and pre-tensioner) (Campbell et al., 2016). The most effective technique is to provide vibrations through the seat, which is the widest space that the driver is in physical contact with (Ji et al., 2011). Petermeijer et al. (2017) highlighted that a visual and auditory warning may not be a highly effective TOR signal in the automated driving mode and evaluated the accuracy of driver response against a stimulus provided by the haptic seat. The results indicated that although the vibrations could effectively facilitate a takeover, the information carried by the vibrations (direction signal) could not be accurately recognized by the driver. Consequently, in addition to a haptic seat, other display signals should be used to impart accurate information. Telpaz et al. (2015) considered the effects of a haptic seat by comparing the scenarios involving and not involving a haptic warning. The experimental group that experienced a haptic seat warning exhibited a smaller reaction time.

The effects of visual, auditory, and haptic modalities and a variety of their combinations have been compared. Melcher et al. (2015) employed a cluster display, mobile phone, acoustic gong, and brake jerk as the HMI components. The effects of the use of the mobile phone and brake jerk were compared to those of a cluster, and an acoustic warning was considered as the baseline during the operation of a non-driving related task. The combined use of a mobile phone and brake jerk resulted in the lowest TOR reaction time. Borojeni et al. (2016) presented the results of a survey research and driving simulator experiment. Two LED lights on the steering wheel were used to provide the visual warning, beep sounds were provided as an auditory warning, and a haptic seat was used. The fastest takeover corresponded to a combination of both the visual and auditory warnings.

Most of the existing studies indicated that a TOR warning using visual, auditory, and haptic modalities was highly effective. Specifically, to deliver a high-priority message to the driver, the use of a multimodal warning was recommended to reduce the information processing time. This recommendation is likely to be reflected in the mass production process of automobiles. BMW reported the use of a visual and auditory modality as a TOR warning to ensure a successful takeover between a Level 3 vehicle and driver (BMW, 2020). Consequently, multimodal warnings are recommended for and currently in use in conditionally automated vehicles; however, the research to clarify the effectiveness of the HMI components integrated in multimodal warnings remains limited. To overcome these limitations, in this study, multimodal warnings involving the previously studied HMI components (visual: cluster, cluster + LED light, auditory: earcon, speech message, haptic: haptic seat) were formulated, and the data obtained in a human-in-the-loop experiment in a driving simulator were analyzed to evaluate the warning performance. This work is similar to the previous studies in that the performance of TOR warnings is analyzed using the same indicators, such as the perception time, reaction time, and vehicle control of the driver. Nevertheless, the contents and results of this study pertain to an analysis of original multimodal warning designs, and the presented findings are expected to have guiding significance for the design of automated driving systems.

The following hypotheses were considered to verify the effects of the visual, auditory, and haptic seat warnings to signal a TOR. We used the name (i.e., the generally used term) of the independent variables (i.e., the warnings) when we were building the hypothesis; however, the reader should consider them design factors.

H₁: The perception time, reaction time, steering deviation, and/or SCR amplitude differ with the type of visual warning, i.e., cluster warning only vs LED lights installed on A-pillar with the cluster warning.

H₂: The perception time, reaction time, steering deviation, and/or SCR amplitude differ with the type of auditory warning, i.e., earcon and speech message.

H₃: The perception time, reaction time, steering deviation, and/or SCR amplitude differ with haptic warnings, i.e., with and without seat vibration.

H₄: Interaction effects occur among different designs of visual, auditory, and haptic warnings.

3. Method

3.1. Apparatus

A driving simulator (KMU-DS) corresponding to a Hyundai LF SONATA (2016) cabin was used in the human-in-the-loop experiment (Fig. 1). A visual warning was provided in the form of a combination of an LED strip on the A-pillar and an icon appearing on the cluster. An auditory warning was provided using three speakers. A driver seat equipped with nine vibration motors was used to provide



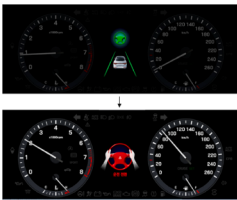
Fig. 1. Experimental equipment: KMU-driving simulator equipped with the multimodal warning system.

a haptic warning. SCANer studio ver. 1.8 of AV Simulation was used to create a certain driving environment reflecting variations in the events and traffic. The Smart Eye Pro-3cam system of Smart Eye was employed; it tracked the position of the driver's gaze before and after the alert had informed him or her about the takeover. The GSR data were acquired with BIOPAC MP160. The temperature of the experimental environment was approximately 73.4 °F (23 °C); the simulator was installed in an area that was not largely affected by changes in the atmospheric temperature and humidity.

3.2. Material (Warning modality design)

In this study, we used A-pillar LED light + cluster icon and cluster icon only as the visual warning modality (2 levels), earcon and speech message as the auditory warning modality (2 levels), and seat vibration and no vibration as the haptic warning modality (2 levels). In a previous study, Yun and Yang (2020) used a HUD icon (red color, circular icon, diameter: 19 mm, flash frequency: 3–4 Hz) as the visual modality, earcon as the auditory warning modality, and seat vibration as haptic warning modality. The specifications of earcon and seat vibration were the same. However, the multimodal warning system used herein and that used in the study of Yun and Yang (2020) were different. In other words, the multimodal warning combination method was different. In this study, we compared the effect of eight multimodal warnings by combining A-pillar LED light, cluster icon, earcon, speech message, seat vibration, and no vibration. Yun and Yang (2020) compared the effects of the following combinations of warning modalities: visual, auditory, haptic,

Table 1
Design of visual, auditory, and haptic warnings.

Modality	Warning design	Contents	Frequency (Hz)	Intensity of warning	Description
Visual	A-pillar + cluster (A)		2.86	1,500 [lx] (2.10 [lx] inside the simulator)	Moving cue activated in case of a TOR Color: R:255, G:0, B:0 (C:0, M:100, Y:100, K:0) Length / width of LED strip: 500 mm / 10 mm No. of lights: 30 ea Diameter of LED lens: $\varnothing$ 5 mm
	Cluster-only (C)		N/A	9.30 [lx]	A green icon shown during the autonomous driving mode, which transforms to red when TOR is issued
					
Auditory	Earcon (E)	Mixed tone of Tesla firmware 8.0 and IONIQ of Hyundai	880 + 1760 217	70 [dB]	Repeated until the operation of DDT fallback
	Speech message (S)	"Drive yourself" in Korean			
Haptic	Haptic (H)	Vibration through the back support and seat pan	28–34.5	2 [g]	Vibrations repeated: on for 0.2 s and off for 0.05 s after a TOR
	None (N)	No vibrations through the driver seat	N/A	N/A	N/A

visual-auditory, visual-haptic, audio-haptic, visual-auditory-haptic, and visual-auditory-haptic. The detailed designs of the multimodal warnings considered in this study are summarized in Table 1, and the design method of each modality is described in the following subsections.

3.2.1. Visual-modality warning design

The LED lights on the A-pillar and an icon on the cluster were used as the visual-modality warning designs. The LED lights installed on the A-pillar were expected to stimulate the driver's peripheral vision, as in the study of [Borojeni et al. \(2016\)](#), where a moving cue on the A-pillar was adopted to issue TORs. Especially, in level 3 automated driving, it is recommended to use a high-level warning to induce quick recognition. Although the driver need not monitor the driving situation, they are expected to be prepared to serve as the fallback and achieve a minimal risk condition ([SAE J3016, 2018](#)). For this reason, we used a red-colored light (R:255, G:0, B:0) as the A-pillar LED light ([Campbell et al., 2016; Yun and Yang, 2020](#)). Furthermore, based on the results of the pilot experiment, an LED light with an intensity of 1,500 lx (which is considered relatively high) was used in the experiment. In [ISO 15008 \(2017\)](#), the brightness range of the display is presented as follows—Night condition: approximately 10 lx, twilight condition: approximately 250 lx, day condition: approximately 5,000 lx, direct sunlight condition: approximately 45,000 lx. However, considering the experimental environment in this study (2.1 lx) and the physical area of the lights (Length/width of LED strip: 500 mm/10 mm, number of lights: 30 ea, diameter of LED lens: $\varnothing 5$ mm), it was difficult to apply the standard brightness of the display to the A-pillar LED lights. On the cluster, a green icon designed by the Korea Automotive Technology Institute HMI group was displayed during the automated driving mode, which transformed to red with certain text to indicate a TOR. In a previous study ([Yun and Yang, 2020](#)), a red icon that flashed with a frequency of 3–4 Hz was displayed on the HUD, which corresponded to a lower reaction time compared to other modality designs. Thus, in this follow-up study, we adopted a more visible visual design. The following two visual warning designs were considered: (1) cluster only warning, and (2) LED lights installed on the A-pillar along with the cluster warning, to compare the effects of the A-pillar light warnings.

3.2.2. Auditory-modality warning design

[Yun and Yang \(2020\)](#) studied only an earcon as the auditory modality, and thus, it was necessary to investigate additional auditory warning designs. In this study, the earcon and speech message designed by Jeon's research group were investigated as the auditory warning ([Jeon, 2019; Richie et al., 2018; Hong et al., 2020](#)). Earcon refers to an auditory warning consisting of a sound at a constant frequency, deemed effective in attracting the attention of a driver ([Campbell et al., 2016](#)). However, speech messages can convey relatively more complex and specific information than an earcon. In this study, a mixed signal of 880 Hz and 1,760 Hz was used as an earcon, and a speech signal in which a female speaker said "Drive yourself" was used as a speech message.

3.2.3. Haptic-modality warning design

[Yun and Yang \(2020\)](#) demonstrated the effectiveness of the vibration seat in reducing the reaction time and lane change time, and thus, the haptic design was extensively investigated in the present study. As a haptic warning signal, vibrations were provided through the driver seat. Vibrations with a magnitude of 2 g were repeated for 0.2 s in an interval of 0.05 s ([Fitch et al., 2011](#)), as indicated in Table 1.

3.3. Independent (IVs) and dependent variables (DVs)

The IVs in this study were as follows: 1) visual warning (IV₁), 2) auditory warning (IV₂), and 3) haptic warning (IV₃). IV₁ involved two levels of A-pillar LED lights + cluster warning vs cluster warning only. IV₂ involved two levels of earcon vs speech message. IV₃ involved two levels of seat vibration vs no vibration.

The following DVs were considered to evaluate the performance of the TOR warnings: perception time (PT) and reaction time (RT), in terms of the user behavior; maximum acceleration (MA) and steering wheel angle deviation (SWD), in terms of the vehicle control; and SCR amplitude, as the physiological indicator, as indicated in Table 2.

PT refers to the time between the activation of a TOR warning and the moment at which a human driver looks at the road scene

Table 2
Dependent variables (DVs).

DV	Definition	Purpose	Unit
Perception Time (PT)	Time required for the driver to look at the road ahead after the TOR warning	To evaluate the driver's perception speed of the TOR (Baek et al., 2019)	[s]
Reaction Time (RT)	Time required for the driver to takeover after the TOR warning	To evaluate the reaction speed to perform DDT fallback (Gold et al., 2013)	[s]
Maximum Acceleration (MA)	Square root of the addition of the squared values of the longitudinal and lateral acceleration 2 s after the takeover	To determine the driver's transient control quality of the vehicle after switching to manual driving (Gold et al., 2013)	[m/s ²]
Steering wheel angle deviation (SWD)	Maximum gap in the angles of the steering wheel 2 s after the takeover	To identify the driver's lateral control quality of the steering wheel after the transition to manual driving (Dukic et al., 2006)	[°]
SCR amplitude (SCR amplitude)	The gap between the baseline and maximum of the conductance values after a TOR	To assess the intensity of a stimulus and level of stress triggered by a TOR warning (Conway et al., 2013)	[μ S]

ahead. (Baek et al., 2019). This indicator was employed to examine if the driver intuitively understood the TOR warning.

RT refers to the time required for a human driver to control the steering wheel, brake pedal, or gas pedal upon a TOR warning. This indicator was adopted to determine whether a TOR warning induced a fast reaction from the driver.

MA is defined as the square root of the addition of the squared values of the longitudinal and lateral acceleration (Gold et al., 2013). MA can be used to evaluate the driver's vehicle control quality. In this study, the MA was analyzed for two seconds of manual driving immediately after the takeover. A smaller MA corresponded to a more stable performance of the driver in taking over vehicle control.

SWD is defined based on the maximum gap of the angles of the steering wheel at a specific section under manual driving (Dukic et al., 2006). In this study, this indicator was used to assess the driver's lateral control of the vehicle for two seconds of manual driving immediately after the takeover. A smaller deviation of the steering wheel angle was considered to signify a more stable control after the takeover.

The GSR has an electrical characteristic that changes according to stimulation of the human sympathetic nerve. Galvanic skin response [GSR, which is also known as “electrodermal activity” (EDA)] is a stress-related biosignal (Giannakakis et al., 2019). GSR includes tonic components (i.e., the skin conductance level: SCL) and phasic components (i.e., the skin conductance response: SCR). In this study, “the amount of changes”, which was witnessed in the SCR, was analyzed. When a person feels stressed, the moisture content in the skin increases, thereby increasing the SCR. The analyzed SCR amplitude indicates the stress level. More specifically, the SCR amplitude represents the gap between the baseline and conductance maximum (Setz et al., 2010). Yun and Yang (2020) used the SCR to compare the physiological differences witnessed among drivers per combination of TOR warnings. In this study, the SCR amplitude was defined as the gap between the baseline and maximum of the conductance values after a TOR. We analyzed the differences in the values triggered by the different TOR warnings.

3.4. Experimental design

The experiment was designed to investigate a situation in which a takeover was required due to an automated driving system failure during the automated driving mode (SAE J3016, 2018). As this study adopted a within-subject design, in which a volunteer experienced all the TOR warning combinations, the drivers could develop the learning effect. This effect was minimized by adopting the following experimental design. First, two additional scenarios without a TOR were implemented to prevent the drivers from anticipating a takeover. Thus, all volunteers experienced ten scenarios: eight scenarios involving a TOR (2 visual warnings $\times$ 2 auditory warnings $\times$ 2 haptic warnings) and two scenarios without a TOR, as indicated in Table 3. Second, the timing of TOR in each scenario was randomized through an exponential distribution. Third, we applied a Latin square design (10 $\times$ 10) on ten scenarios and ten TOR timings (Wan and Wu, 2018). This framework enabled each volunteer to experience scenarios with different TOR time points in a random order. For example, volunteer 1 experienced the AEH warning 122.34 s after the start of the first scenario, whereas volunteer 2 experienced the AEH warning 200.78 s after the start of the fifth scenario.

3.5. TOR scenario

All the scenarios were designed with the following common aspects: The vehicle moved autonomously at a speed of approximately 85 kph in the third lane. To prevent a volunteer from looking ahead, a surrogate relative task was assigned to induce a visual load (ISO/TS 14198, 2019). SuRT is one type of many NDRTs (Naujoks et al., 2018); the volunteers are asked to identify the circle with a different size among many circles with identical sizes. The NDRT is performed until a TOR warning is generated; in this moment, the volunteers are expected to stop the SuRT and maneuver the gas pedal, brake pedal, or steering wheel, thereby switching from automated to manual driving. To let the elevated GSR signal recover to the baseline after the TOR, the scenario ended when the volunteer had manually operated the vehicle for 30 s after the takeover (Fig. 2) (Gautam et al., 2018). All of the scenarios were designed with the following common aspects: The vehicle moved autonomously at a speed of approximately 85 kph in the third lane of a four-lane road in the Singal–Osan IC (highway) in South Korea. It was daytime, and the weather was clear. The traffic volume was LOS C (meaning that more than 10 vehicles but no more than 14 vehicles drive per kilometer). To prevent a volunteer from looking ahead, a surrogate relative task was assigned to induce a visual load (ISO/TS 14198, 2019). SuRT is one of the many types of NDRTs (Naujoks et al.,

Table 3
TOR warning modality combinations.

Scenario No.	TOR warning combination	Modality		
		Visual	Auditory	Haptic
1	AEH	A-pillar + cluster (A)	Earcon (E)	Haptic (H)
2	AEN	A-pillar + cluster (A)	Earcon (E)	None (N)
3	ASH	A-pillar + cluster (A)	Speech message (S)	Haptic (H)
4	ASN	A-pillar + cluster (A)	Speech message (S)	None (N)
5	CEH	Cluster only (C)	Earcon (E)	Haptic (H)
6	CEN	Cluster only (C)	Earcon (E)	None (N)
7	CSH	Cluster only (C)	Speech message (S)	Haptic (H)
8	CSN	Cluster only (C)	Speech message (S)	None (N)
9	No TOR	None	None	None
10	No TOR	None	None	None

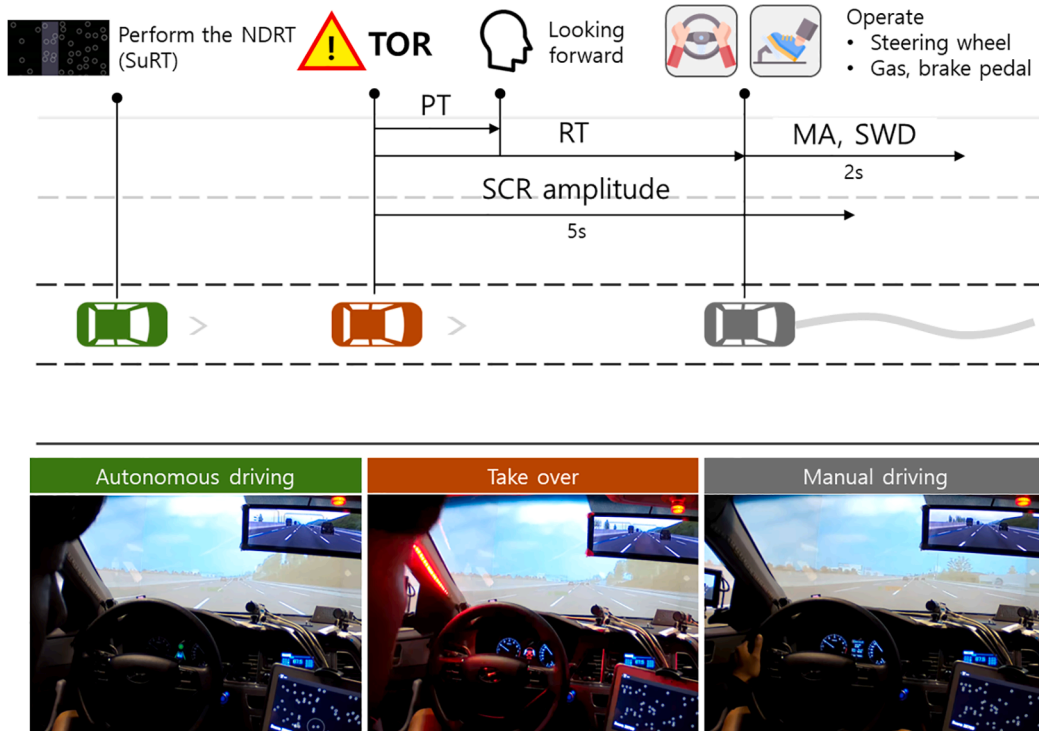


Fig. 2. TOR scenario.

2018); the volunteers were asked to identify a circle of a different size among many circles of the same size. The NDRT was performed until a TOR warning was generated. During automated driving, when ADS system failure occurred, TOR was provided to the driver (unplanned ODD exit). At this moment, the volunteers were expected to stop the SuRT and operate the gas pedal, brake pedal, or steering wheel, thereby switching from automated to manual driving. To allow the elevated GSR signal recover to the baseline after the TOR, the scenario was ended when the volunteer had manually operated the vehicle for 30 s after takeover (Fig. 2) (Gautam et al., 2018). There were no other driving events. The level of difficulty of the 10 scenarios used in this study was the same because the traffic volumes were the same, and the TOR was implemented only in a straight line.

3.6. Volunteers

Thirty-six volunteers, who had a driving experience of more than two years and drove two or three times a week, were recruited. The volunteers included 22 men (61.1 %) and 14 women (38.9 %), with an average age of 40.9 years ($SD = 14.8$) and average driving experience of 170.7 months ($SD = 135.0$). This study was conducted upon approval from the Kookmin University Institutional Review Board and complied with the associated regulations (KMU-201803-HR-174).

3.7. Experimental process

First, the details of the experiment were explained to the volunteers, and the volunteers were requested to fill out informed consent forms. Next, the volunteers were equipped with physiological signal acquisition device(s) to obtain the GSR data, and their eye gaze was calibrated to the eye tracker. After the equipment setting was well-calibrated, the volunteers were requested to perform a test drive to familiarize themselves with the driving simulator and expected task, i.e., takeover sequence. When the test drive began, the car was automatically driven. After 30 s, a TOR was initiated and presented to a volunteer; he or she maneuvered the steering wheel, brakes, or pedal to control the vehicle. The scenario ended after another 30 s. When a volunteer was considered unfamiliar with how to take control of the vehicle, he or she performed another test drive. The warning type used in this test driver is a beep sound, which was not included in actual experiments. After the test drive, the main experiment was initiated, which involved ten 3–5 min scenarios implemented in a random order (In short, different types of TORs were used for each experiment.). The volunteers took a break after running the scenario five times to reduce fatigue. The duration of the break varied depending on the physical conditions of the volunteers. After they had engaged in the scenario five times, the main experiment was completed. Upon the completion of the experiment, the moderator of the experiment checked whether the data had been acquired and secured and officially ended the experiment. The total time for the experiment was approximately two hours per person.

4. Results

The average and standard deviation values of the DV for each multimodal combination are presented in Table 4. A three-way analysis of variance with repeated measures was performed to analyze the main and interaction effects of the TOR warnings. The significance level was set as $\alpha = 0.05$. No order/learning effect was found.

4.1. Main effects of the visual (H_1), auditory (H_2), and haptic modality (H_3) warning

4.1.1. Main effect of H_1

The average RT when the A-pillar LED lights were implemented was 2.27 s, significantly lower than that without the A-pillar LED lights, 2.51 s ($p = 0.022$) (Tables 4 and 5 and Fig. 3). The average SWD with the lights was 8.24 s, significantly higher than that without the lights (6.44 s) ($p = 0.024$). No significant difference was observed for PT, MA, and SCR amplitude. Overall, the RT and steering deviation differed between the cases with only the cluster warning and with LED lights installed on the A-pillar with the cluster warning.

4.1.2. Main effect of H_2

The average SWD when the earcon was used was 6.66, significantly smaller than the average value (8.02) when the speech message was used ($p = 0.044$) (Table 4 and 5; Fig. 3). However, no main effects were observed in the case of PT, RT, MA, and SCR amplitude when the auditory modality warning was used. Overall, differences between the earcon and speech message cases were noted only in terms of the steering deviation.

Table 4

Mean and standard deviation (SD) of dependent variables.

Modality	Warning	Mean (SD) of dependent variables				
		PT [s]	RT [s]	MA [m/s ²]	SWD [°]	SCR amplitude [μS]
Visual (V)	A	0.92 (0.72)	2.27 (1.13)	1.94 (1.89)	8.24 (7.43)	2.96 (2.61)
	C	1.05 (0.62)	2.51 (1.11)	1.81 (1.92)	6.44 (6.30)	2.77 (2.65)
Auditory (A)	E	0.95 (0.72)	2.34 (1.17)	1.91 (1.92)	6.66 (6.05)	2.98 (2.74)
	S	1.01 (0.62)	2.45 (1.08)	1.84 (1.89)	8.02 (7.68)	2.75 (2.52)
Haptic (H)	H	0.96 (0.52)	2.21 (1.01)	1.99 (1.98)	6.97 (6.26)	3.21 (2.47)
	N	1.00 (0.80)	2.58 (1.20)	1.76 (1.83)	7.71 (7.55)	2.53 (2.75)

Table 5

Repeated measures ANOVA results (df = 1 for all indicators), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Modality	PT		RT		MA		SWD		SCR amplitude	
	F	p	F	p	F	p	F	p	F	p
Visual (H_1)	4.100	0.051	5.738	0.022*	1.041	0.315	5.548	0.024*	0.968	0.332
Auditory (H_2)	0.630	0.433	0.865	0.359	0.271	0.606	4.366	0.044*	1.189	0.283
Haptic (H_3)	0.365	0.550	21.802	<0.001***	2.001	0.166	1.622	0.211	12.643	0.001**
V × A (H_4)	0.827	0.369	1.015	0.321	1.874	0.180	16.990	<0.001***	6.081	0.019*
V × H (H_4)	0.857	0.361	0.520	0.476	0.013	0.911	2.184	0.148	0.001	0.974
A × H (H_4)	0.023	0.880	0.349	0.559	0.269	0.607	0.001	0.974	0.263	0.611
V × A × H (H_4)	0.369	0.547	3.513	0.069	0.015	0.904	3.301	0.078	0.609	0.440

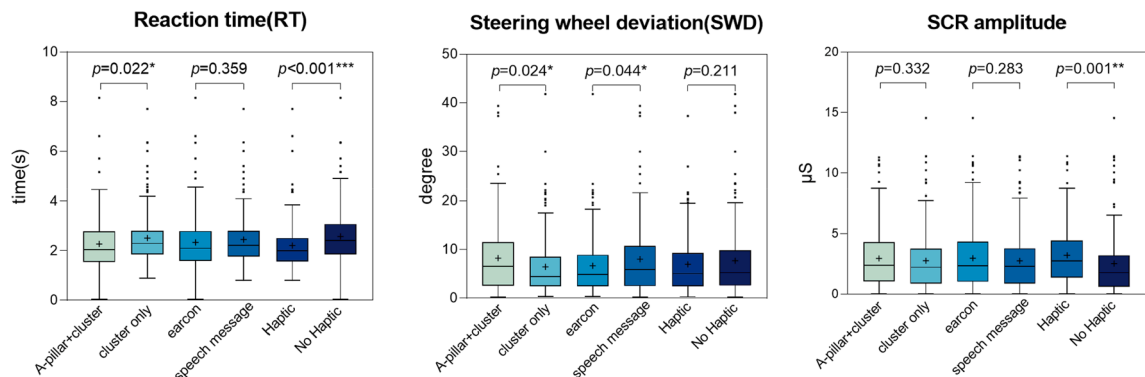


Fig. 3. Main effects (visual, auditory, and haptic modalities).

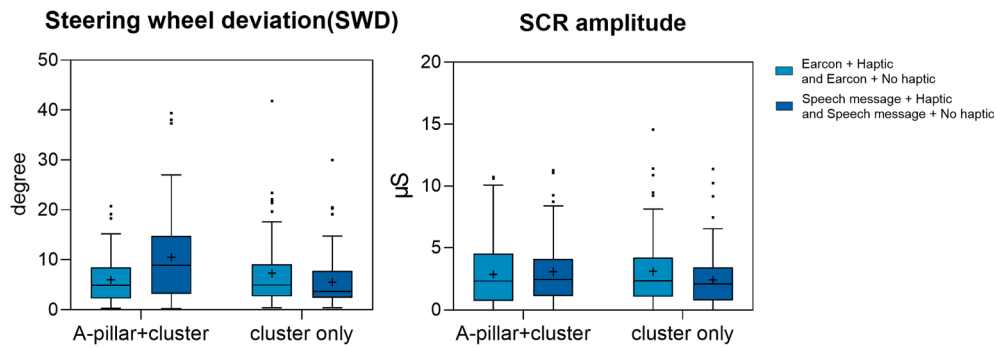


Fig. 4. Two-way interaction effects (visual × auditory modality warning).

4.1.3. Main effect of H_3

The average RT upon a TOR warning was 2.21 s when the haptic seat was used, significantly lower than the average value (2.58 s) without the seat ($p < 0.001$), as indicated in Tables 4 and 5 and Fig. 3. The average SCR amplitude when the haptic seat was used was 3.21 μS , significantly higher than that (2.53 μS) without the seat ($p = 0.001$). No main effects were found in terms of the PT, MA, and SWD upon the activation of the haptic modality warning. Overall, differences in the cases with and without the seat vibrations existed only in terms of the RT and SCR amplitude.

4.2. Interaction effects of the visual, auditory, and haptic modality warning (H_4)

The experiments of the visual–auditory modality exhibited significant interaction effects in terms of the SWD and SCR amplitude ($p < 0.001$ and $p = 0.019$, respectively) (Table 5 and Fig. 4). The average SWD of the A-pillar–earcon warning (AE) was 6.01° , which is lower than that (10.46°) of the A-pillar–speech message warning (AS). Furthermore, the average SWD of the cluster–earcon warning (CE) was 7.31° , which is higher than that (5.58°) of the cluster–speech message (CS). The average SCR amplitude of the AE warning was 2.86 μS , which is lower than that (3.06 μS) of the AS. However, the average SCR amplitude (3.10 μS) of the CE was higher (2.44 μS) than that of the CS. In addition, the visual–auditory modality showed no interaction effect in the PT, RT, and MA indicators ($p = 0.369$, 0.321, and 0.180). The visual–haptic warnings showed no interaction effects for all indicators ($p = 0.361$, 0.479, 0.911, 0.148, and 0.974). Likewise, the auditory–haptic and visual–auditory–haptic warnings did not exhibit interaction effects for all indicators [$(p = 0.880, 0.559, 0.607, 0.974, \text{ and } 0.611)$ and $(p = 0.547, 0.069, 0.904, 0.078, \text{ and } 0.440)$, respectively] (Table 5). In summary, interaction effects occurred only between visual and auditory warnings.

5. Discussion

The implications of the results described in Section 4 can be elaborated as follows. This study compares the effects of A-pillar LED light, earcons, speech messages, and haptic seat warnings used to signal a TOR; the results originate from experiments done on a total of 36 volunteers. A three-way ANOVA was performed for statistical analysis to compare the main and interaction effects of TOR warnings. The results indicated that the visual, auditory, and haptic warnings demonstrated the main effect(s). The LED light manifested a significantly faster RT ($p = 0.022$) and a significantly larger SWD ($p = 0.024$) compared to the cluster. Moreover, the earcon resulted in a significantly smaller SWD than the speech message ($p = 0.044$). The haptic seat led to a significantly faster RT ($p < 0.001$) and a significantly larger SCR amplitude ($p = 0.001$) compared to None (No haptic). The visual and auditory modalities exhibited interaction effect(s). The earcon resulted in a small SWD ($p < 0.001$) and skin conductance response amplitude (SCR) ($p = 0.019$) when combined with the use of LED lights on the A-pillar. By contrast, the speech message without LED lights led to a small SWD and SCR amplitude.

Visual modality warning: The RT and steering deviation exhibited a statistically significant difference between the case involving only a cluster warning and that with LED lights installed on the A-pillar with the cluster warning. It can be considered that the A-pillar LED lights prompted the driver to react faster but resulted in an unstable lateral control of the vehicle. Notification that stimulates peripheral vision inducing a quick response is the same as the findings of Borojeni et al. (2016) and Çapalar and Olaverri-Monreal (2017). On the other hand, the lateral control was unstable, and it can be attributed to large steering wheel deviation by the driver distracted by LED light experience (Kountouriotis and Merat, 2016) or the driver surprised by the use of higher brightness than necessary in this study. In the future, it is necessary to develop a TOR warning with an appropriate illuminance.

Auditory modality warning: The steering deviation was statistically different between the cases involving an earcon and a speech message. Such as for the visual modality, the lateral control triggered by the speech message was more unstable than the earcon; this indicates that the driver may have been distracted. Compared with earcons, speech messages can add mental workload; nevertheless, they can evoke positive emotions. Jeon (2019) explained that a speech message is more pleasing than an earcon. On the other hand, Forster et al. (2017) demonstrated that speech + generic warnings result in a faster reaction time than their generic counterpart. However, the results of this study revealed that there was no significant difference in the reaction times between earcons (similar to the

generic signal in that it repeats a certain tone) and the speech message. It can be assumed that combinations of speech and generic warning signals result in different effects. In the future, the effects of different combinations of warning types, such as speech + generic warnings, on drivers and their performance should be investigated and compared.

Haptic modality warning: The RT and SCR amplitude exhibited a statistically significant difference between the cases with and without seat vibrations. It can be considered that the provision of vibrations through the seat allowed the drivers to react faster but induced a certain stimulus and stress. Although the fact that it reduced the driver's reaction time is the same as the report by [Telpaz et al. \(2015\)](#), this study verified the results of previous studies by comparing the effects of the presence of a haptic seat used in a multimodal warning. On the other hand, the SCR amplitude was high when the haptic seat was used, and it was different from the result that the SCR onset time of the visual warning was smaller than that of other multi-modal warnings in the study of [Yun and Yang \(2020\)](#). The difference is that while [Yun and Yang \(2020\)](#) used only HUD as visual modality and earcon as auditory modality, this study compared the effects of haptic modality using A-pillar LED light as visual modality and earcon and speech message as auditory cluster modality.

Interaction effect between visual and auditory warning: Compared to the AS, the AE warning enabled a more stable lateral control of the vehicle and induced a lower level of stress for the drivers. Moreover, the CE warning led to a more unstable lateral control than the CS. No interaction effects triggered by the visual-haptic, auditory-haptic, and visual-auditory-haptic warnings were observed. We assumed that almost all modalities would generate interaction effects when we designed the experiment. However, only the visual–auditory modality displayed interaction effects. According to the interviews with the volunteers, a cognitive load likely affected the interaction effect, as described in the following paragraph.

Volunteer 6: The warning with LED lights was the last to be recognized and thus was not helpful in initiating the takeover.

Volunteer 7: The speech message was helpful for me to perceive the situation as it conveyed a clear message, and the haptic seat was useful in inducing a rapid response. However, when the LED light, speech message, and seat vibrations were activated simultaneously, they stressed me out and distracted me from what actually mattered, thereby making things worse.

According to the subjective opinions, the split-attention effect indicated that a cognitive load likely occurred in the visual working memory when both visual and auditory data were simultaneously provided ([Kalyuga et al., 1999](#)). It can be reasonably inferred that a combination of the A-pillar LED lights and speech message, which required an interpretation of the contents, likely triggered a cognitive load, resulting in physiological stress and unstable lateral control of the vehicle ([Borojeni et al., 2016](#); [He & McCarley, 2011](#)).

The TOR study and the results of this study are compared as follows. Firstly, this study compared the TOR effect using the existing HMI interface design as a multimodal warning. [Çapalar and Olaverri-Monreal \(2017\)](#) used an ambient light system during TOR to induce driver's attention. The reaction time was shorter when the ambient light system was used compared to the scenario without the light system. However, the difference was not statistically significant. The researchers revealed that the use of LED light resulted in a significantly longer reaction time than experiments without LED light. In both studies, the reaction time was shorter when the peripheral vision of the driver was stimulated with light. However, in this study, the result is statistically significant; it may have been caused by the difference in the lighting specifications. For instance, we used red light with 1,500 lx intensity; in addition, motion cue(s) were applied. [Çapalar and Olaverri-Monreal \(2017\)](#) adopted different light colors (i.e., red, blue, and green) with unnoticeably different. In this study, the light stimulated the peripheral vision further, thereby leading to the statistically significant difference in the reaction times. [Forster et al. \(2017\)](#) compared the generic and the combination of generic and speech messages as TORs; according to the results, the subjects preferred speech messages. In the group of volunteers experiencing the generic alert first, the speech + generic warning induced a significantly faster reaction time (pushing buttons, hands-on activation, hands-free) than the generic warning. However, the results of this study revealed that there was no significant difference in the reaction times between earcons (similar to the generic signal in that it repeats a certain tone) and the speech message. The different result is assumed to be generated by the fact that [Forster et al. \(2017\)](#) combined a speech message and generic alert, whereas only speech message was used in this study for auditory modalities. [Petermeijer et al. \(2017\)](#) reported that the notification with both auditory and haptic seats is effective for takeover. As [Petermeijer et al. \(2017\)](#) compared unimodal and multimodal warnings, its design was different from the one of this study (which compared multimodal warnings). One similarity between the two studies is that the multimodal warning combined with the use of a haptic seat reduces the reaction time. The haptic warning did not have any interaction effect with other modalities, which means that the haptic seat can be used to decrease effectively the reaction time for takeovers, regardless of the design specifications of other modalities. This study used multimodal warning, including visual modality, and showed that the use of LED light and the haptic seat affected the driver's reaction time. Next, a study comparing single modalities and their combinations was conducted ([Bazilinskyy et al., 2015](#); [Borojeni et al., 2016](#); [Petermeijer et al., 2017](#); [Yun and Yang, 2020](#)), while this study used only the multimodal warning and compared the takeover effect. Finally, a survey paper suggested that multimodal was effective for takeover ([Naujoks et al., 2019](#); [Morales-Alvarez et al., 2020](#)). This study extended the result and compared the main effect and the interactive effect of multimodal and suggested facts to consider when designing multimodal warning (especially LED light, earcon, and speech message) based on the result.

Limitations: In this study, the various situations in which the automated driving mode is required to be deactivated were not considered. The effects of the warnings may differ depending on the factors contributing to a switch from automated to manual driving, level of urgency that a situation presents, traffic volume, and weather conditions. Additional studies are required to generalize the HMI design that was used in the experiment of this study. We used the color red (R:255, G:0, B:0) for the A-pillar and an LED light with 1,500 lx intensity (i.e., a relatively high intensity) for the experiment. However, red color is similar to the taillights and brake lights of the vehicle in front, which is a limitation because it can confuse drivers. Therefore, it is necessary to study appropriate colors that can be distinguished from other lights having warning properties similar to TOR. Brightness, too, is a limitation in this study. Because the

design specifications are considered too bright for mass production, more studies are needed to identify the suitable brightness levels and color of the LED lights. For example, appropriate brightness could be set considering time zones (night, twilight, day, and direct sunlight conditions) and weather (clear, cloudy, etc.). In this study, we compare the differences in perception time, reaction time, steering deviation, and/or SCR amplitude according to the warning type. However, the brightness of the A-pillar LED light was difference from that of the cluster icon. In a comparison of the differences according to warning types, the brightness level could be the same. Furthermore, the seat vibrated at 2 g vibration intensity for 0.2 s and remained idle for 0.05 s; the differences due to diverse design specifications should be compared in the future. For instance, the following factors of experiments should be compared: differences in the brightness and color of the LED lighting, differences in the sound volumes of earcons, differences in the intensity levels of motor vibrations, and directions of vibrations. In addition, as the experiment for each combination of modalities was repeated, the volunteers experienced them several times; the learning effect may have affected them. We ensured that the sequence of the experiment was counterbalanced and that there was no learning effect after an experiment; nevertheless, the repetition of the experiment may have introduced unexpected effects. Because we selected the SuRT as the NDRT to prevent volunteers from looking ahead during automated driving, the possibility of a longer reaction time due to the visual load must be considered. A comparison of the results obtained using diverse NDRTs that can be employed in level 3 automated driving may yield meaningful results. An *n*-back can induce cognitive loads, and there are everyday NDRTs such as reading or eating. To overcome this limitation, a follow-up research study must be implemented to assess the performance of the warnings in situations in which a planned operational design domain exit (e.g., highway exit) is implemented.

6. Conclusions

This study was aimed at comparing the effectiveness of multimodal—visual, auditory, and haptic—warnings. Two warnings were designed per modality based on the literature review, and eight TOR warning combinations, created by combining these modalities, were evaluated using a driving simulator. We propose the following recommendations pertaining to the use of TOR warnings in an unexpected situation in which a takeover must be implemented.

1. Because A-pillar LED lights can reduce the RT, their use is recommended in situations with a high level of urgency. However, these lights may lead to unstable lateral control after the takeover.
2. Because haptic seats can reduce the RT, their use is recommended in situations involving a high urgency. Such seats may induce a high intensity stimulus or stress to the driver, although this aspect also indicates that a clear warning can be delivered to the driver.
3. A-pillar LED lights should be combined with earcons as auditory warning signals. When A-pillar LED light is unavailable, a speech message should be used as an auditory warning signal. Overall, the effectiveness of multimodal warnings consisting of A-pillar LED lights, earcon, speech messages, and haptic seats was compared and different techniques to utilize these warnings were examined. The results can facilitate the design of warnings in the development phase of conditionally automated vehicles. Follow-up studies must be performed to identify the most effective warning depending on the physical conditions of the driver by monitoring these aspects and clarifying the effectiveness of the warnings in experiments involving an actual car.

CRedit authorship contribution statement

Sara Hong: Investigation, Data curation, Writing – original draft, Visualization. **Ji Hyun Yang:** Supervision, Project administration, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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